

수위 예측 AI 성능 개선 결과 2

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 67개 中 64개 데이터 파일 학습데이터로 사용

검증 : 학습 데이터에서 제외한 2개 데이터 (220713, 230711) + 학습에서 검증에 사용된 데이터 (250716, 250814) 사용

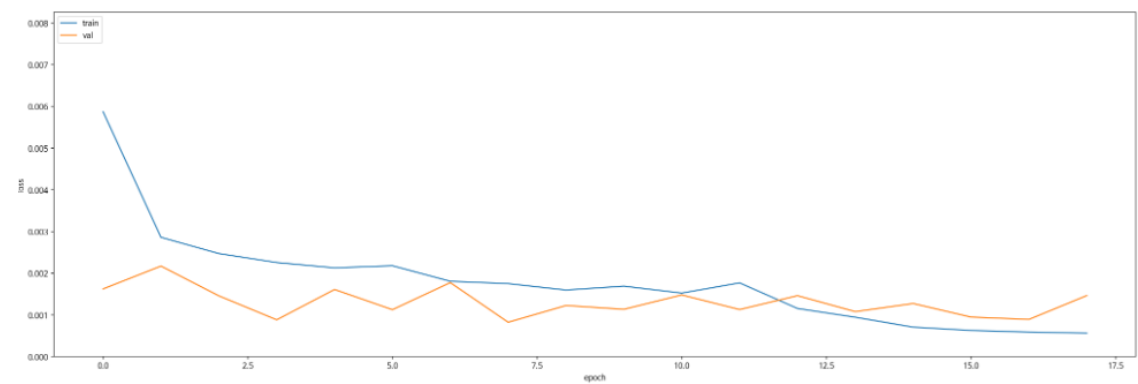
학습데이터 순서 복구,
비율 7:2:1로 변경
(증설 전 후 비교)

loss_function : MSE

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0929	0.0566	0.1681	0.916973	0.918180	32.452990

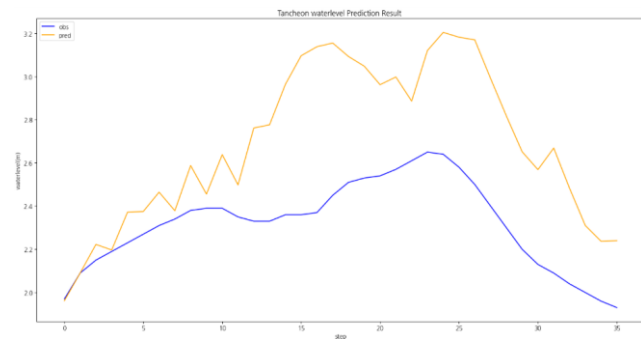
Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0952	0.0712	0.1316	0.885653	0.887989	18.568189

Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0662	0.0498	0.0978	0.908553	0.917067	4.491322

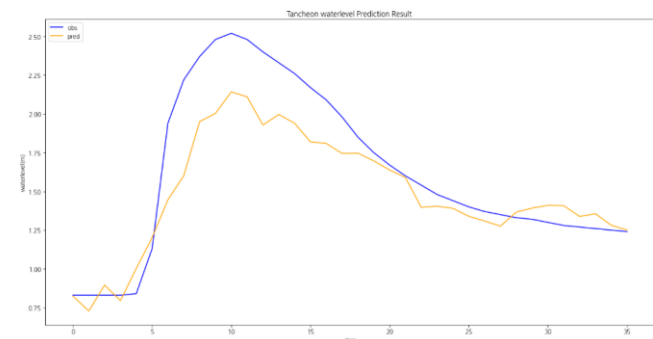
loss_function : MSE

예측 코드 - 궁내교

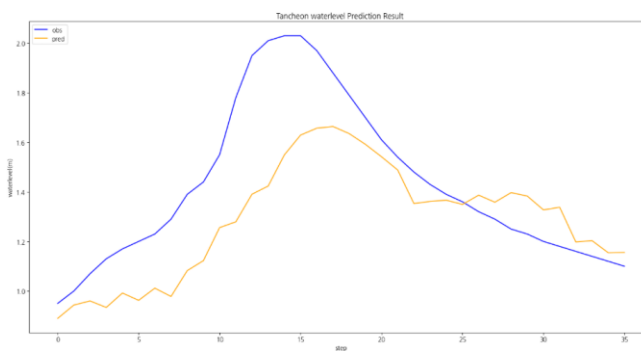
	궁내교
증설 전	testdata_220712 r2 : -3.7460557062787814 mse : 0.18831587329869987 mae : 0.37048792507913375 rmse : 0.43395376861907753
	testdata_230711 r2 : 0.7858148103033311 mse : 0.059937791720316914 mae : 0.17798842834101786 rmse : 0.24482195922816424
증설 후	testdata_250716 r2 : 0.3947407285994061 mse : 0.06072095512751068 mae : 0.19081864602035944 rmse : 0.24641622334479255
	testdata_250814 r2 : -0.056341420985954516 mse : 0.07447427089080348 mae : 0.2094912936952379 rmse : 0.27289974512777304



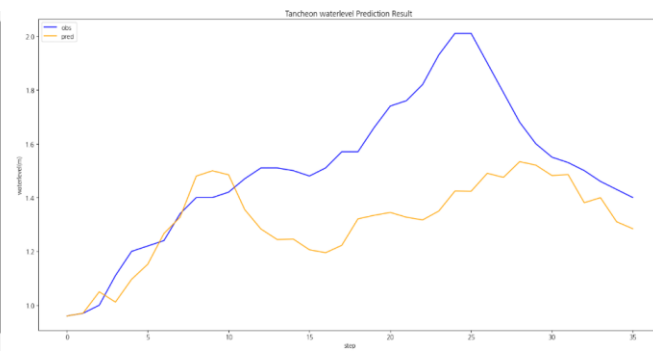
testdata_220712



testdata_230711



testdata_250716



testdata_250814

성능개선 1 최종 결과가 더 나음

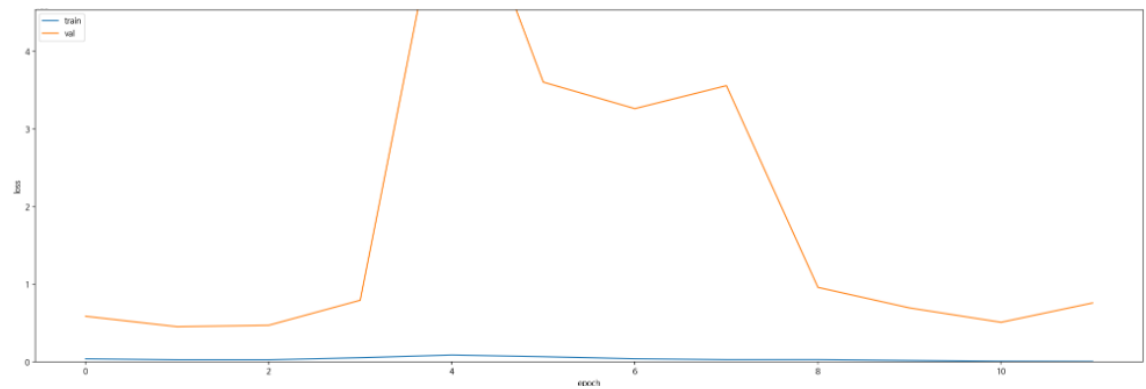
Loss Function : MAPE (%)

loss_function : MAPE

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.7818	0.5205	1.3390	-4.265376	0.061747	126.867669

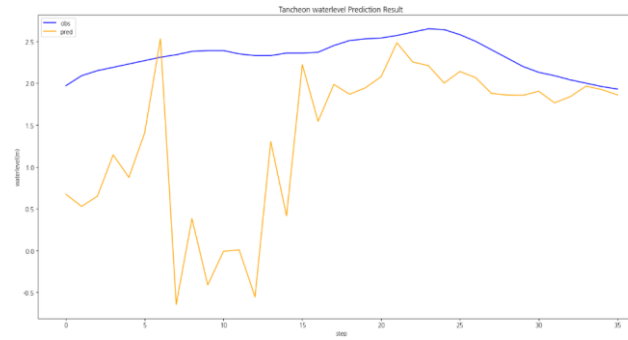
Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.6905	0.5260	1.2014	-8.535100	0.040381	285.320350

Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.6766	0.5380	1.2127	-13.069486	0.029767	383.395654

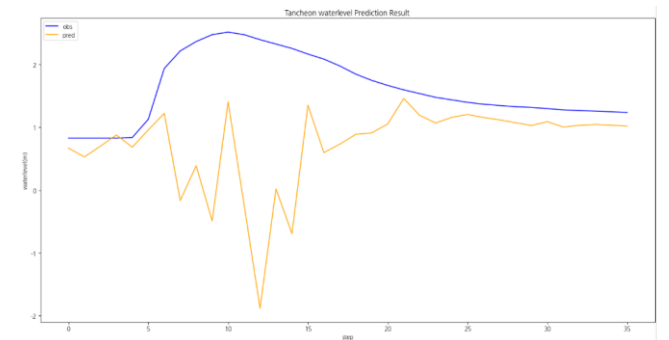
loss_function : MAPE

예측 코드 - 궁내교

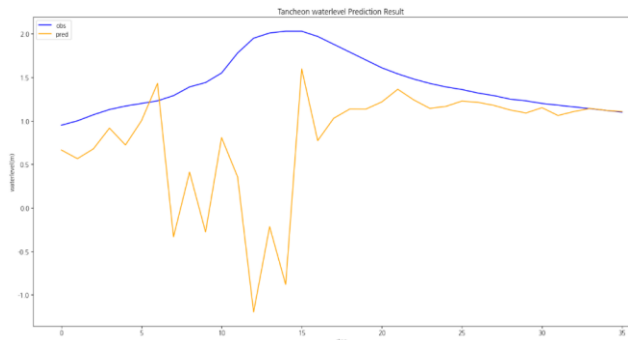
	궁내교
증설 전	testdata_220712 r2 : -40.30121120544763 mse : 1.6387657747376347 mae : 0.9424280889973873 rmse : 1.280142872783204
	testdata_230711 r2 : -5.705601368884521 mse : 1.8765020064033087 mae : 0.890655833081239 rmse : 1.3698547391615319
증설 후	testdata_250716 r2 : -9.178691934208373 mse : 1.0211489941551932 mae : 0.6335310859481492 rmse : 1.0105191706025143
	testdata_250814 r2 : -22.70733371491017 mse : 1.6714164171797394 mae : 0.8478909284538693 rmse : 1.2928327104384927



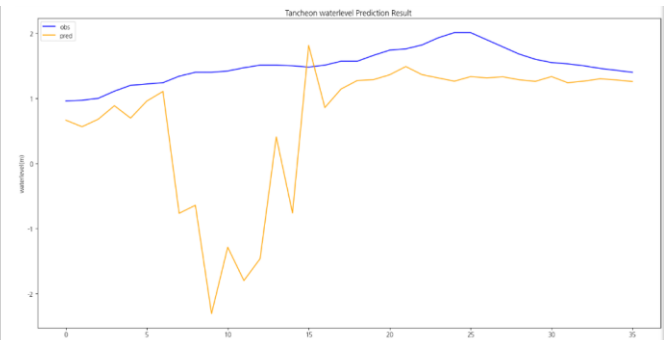
testdata_220712



testdata_230711



testdata_250716



testdata_250814

Loss Function : MAPE (%)
불가능

seq_length : 36 이상 증감 (6시간)

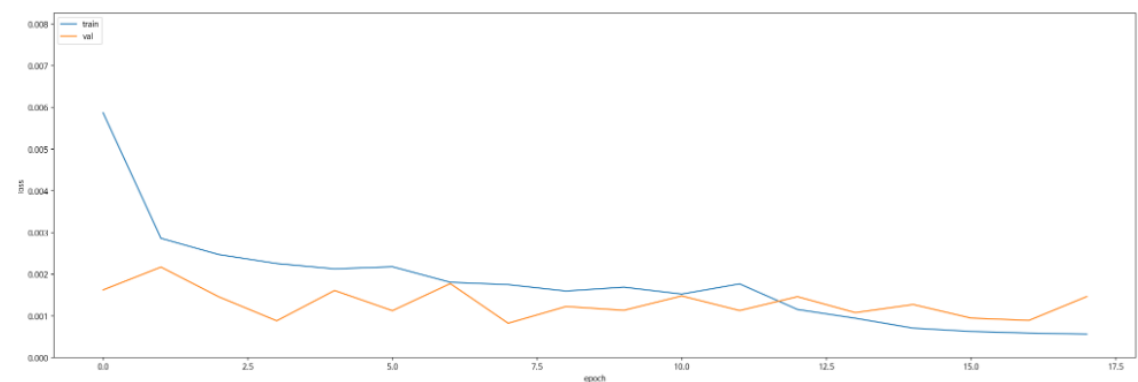
0. 기존 seq_length : 36

seq_length : 36

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

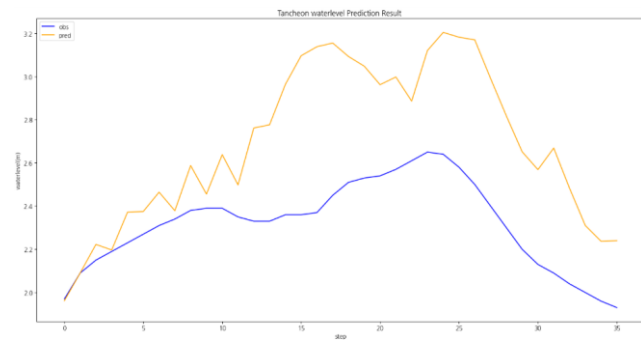
Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0929	0.0566	0.1681	0.916973	0.918180	32.452990

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0952	0.0712	0.1316	0.885653	0.887989	18.568189

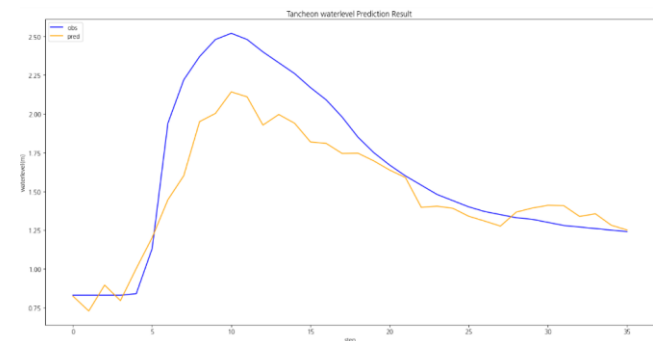
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0662	0.0498	0.0978	0.908553	0.917067	4.491322

seq_length : 36
예측 코드 - 궁내교

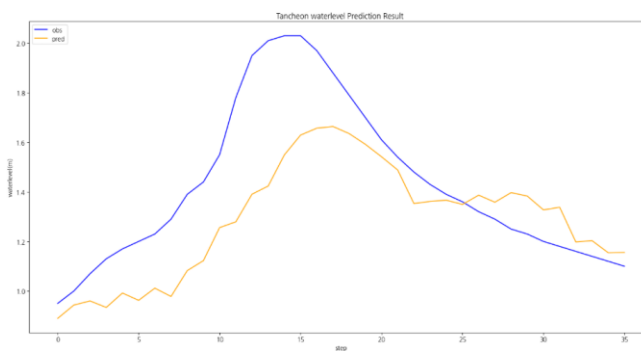
	궁내교
증설 전	testdata_220712 r2 : -3.7460557062787814 mse : 0.18831587329869987 mae : 0.37048792507913375 rmse : 0.43395376861907753
	testdata_230711 r2 : 0.7858148103033311 mse : 0.059937791720316914 mae : 0.17798842834101786 rmse : 0.24482195922816424
증설 후	testdata_250716 r2 : 0.3947407285994061 mse : 0.06072095512751068 mae : 0.19081864602035944 rmse : 0.24641622334479255
	testdata_250814 r2 : -0.056341420985954516 mse : 0.07447427089080348 mae : 0.2094912936952379 rmse : 0.27289974512777304



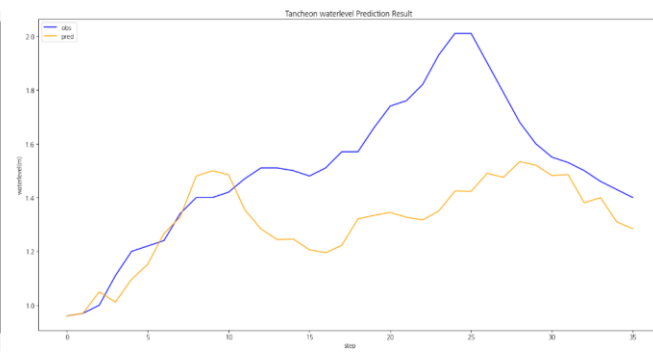
testdata_220712



testdata_230711



testdata_250716



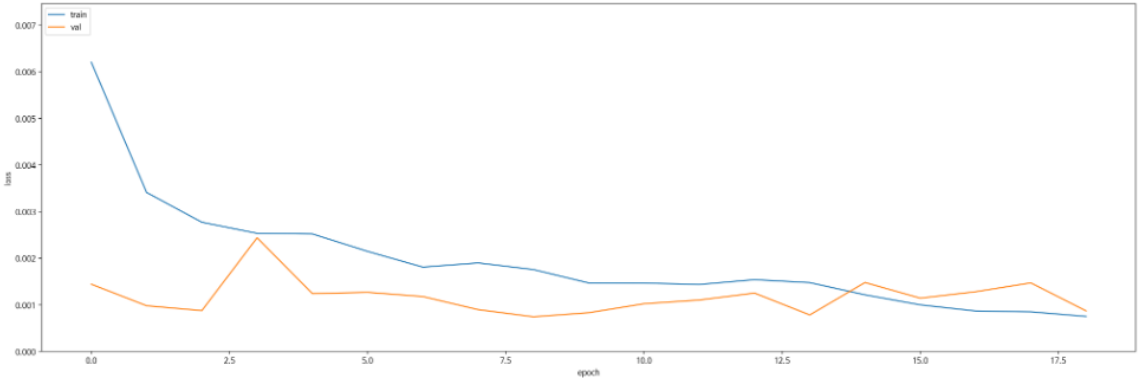
testdata_250814

1. seq_length : 48 (8시간)

seq_length : 48
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

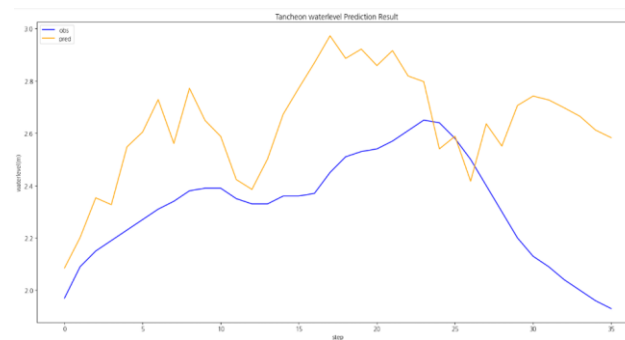
Train (4659 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0986	0.0583	0.1819	0.908400	0.912644	4.648186

Valid (1190 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0934	0.0721	0.1250	0.897810	0.899730	15.198126

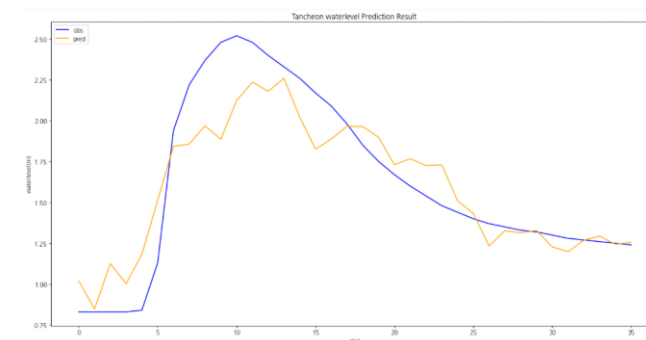
Test (522 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0697	0.0551	0.1148	0.868960	0.870691	93.554015

seq_length : 48
예측 코드 - 궁내교

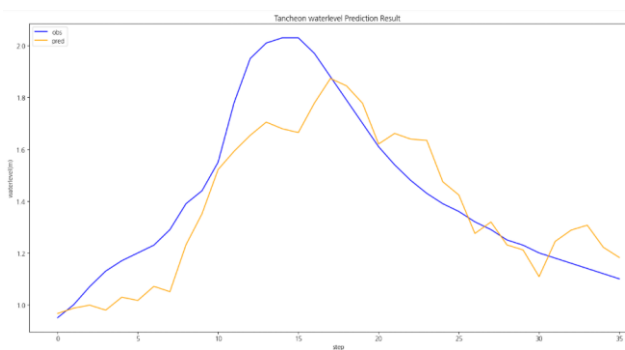
	궁내교
증설 전	testdata_220712 r2 : -2.560335156943706 mse : 0.14126838490937313 mae : 0.32217343224419487 rmse : 0.3758568675830909
	testdata_230711 r2 : 0.825501108212527 mse : 0.0488319395295068 mae : 0.16682389676570894 rmse : 0.22097950024721025
증설 후	testdata_250716 r2 : 0.7542307880110747 mse : 0.024656113500533407 mae : 0.12436577856540677 rmse : 0.15702265282606012
	testdata_250814 r2 : 0.19244131198133563 mse : 0.056934569919249195 mae : 0.185254346927007 rmse : 0.23860966015492582



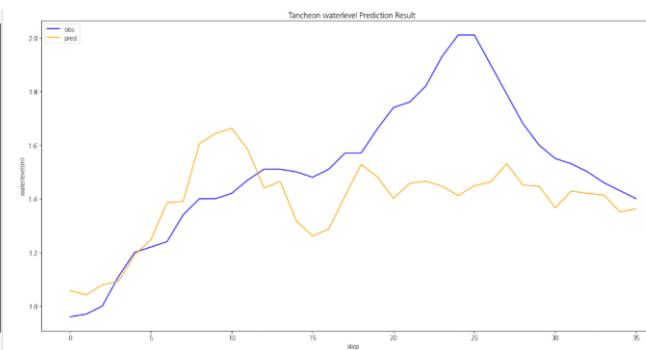
testdata_220712



testdata_230711



testdata_250716



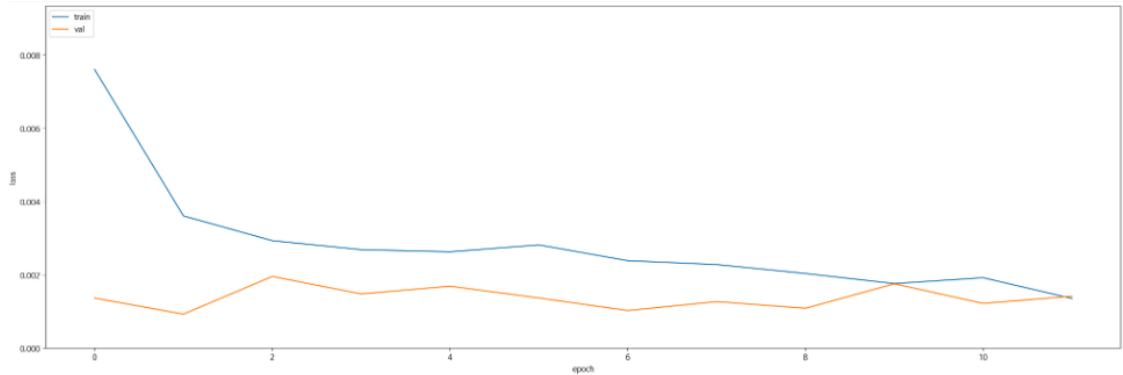
testdata_250814

2. seq_length : 60 (10시간)

seq_length : 60
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

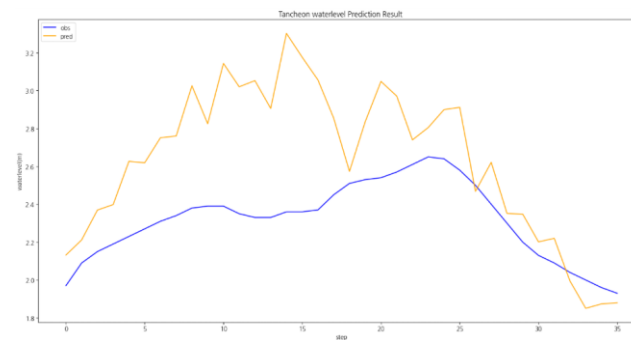
Train (4119 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1138	0.0649	0.2400	0.851486	0.852724	5.146548

Valid (1034 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1051	0.0803	0.1399	0.873851	0.882577	14.866536

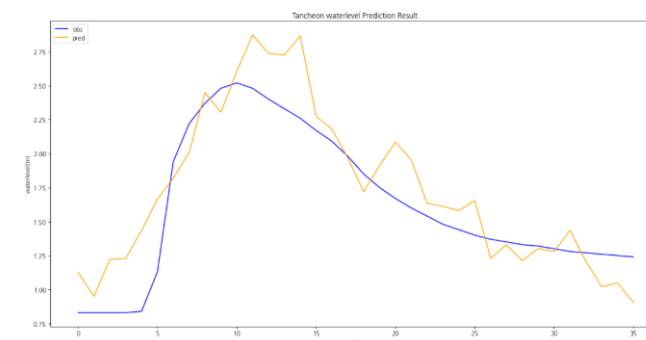
Test (450 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0773	0.0601	0.1125	0.874091	0.881549	10.511493

seq_length : 60
예측 코드 - 궁내교

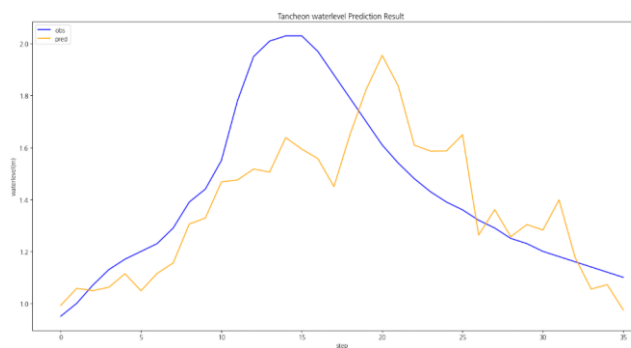
	궁내교
증설 전	testdata_220712 r2 : -3.4182190272963746 mse : 0.1753078400343109 mae : 0.3363218220074972 rmse : 0.4186977908161338
	testdata_230711 r2 : 0.7329701183406903 mse : 0.07472590169592619 mae : 0.22037673400508037 rmse : 0.27336038794222944
증설 후	testdata_250716 r2 : 0.49407557481516895 mse : 0.05075546261104266 mae : 0.17494943426714998 rmse : 0.22528973037189837
	testdata_250814 r2 : 0.5309707340261354 mse : 0.03306754039546155 mae : 0.15624683883455065 rmse : 0.18184482504449104



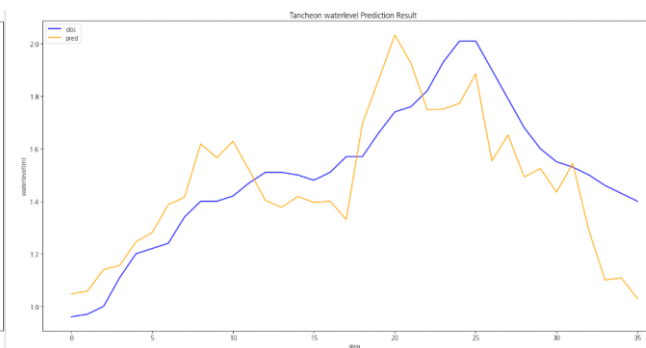
testdata_220712



testdata_230711



testdata_250716



testdata_250814

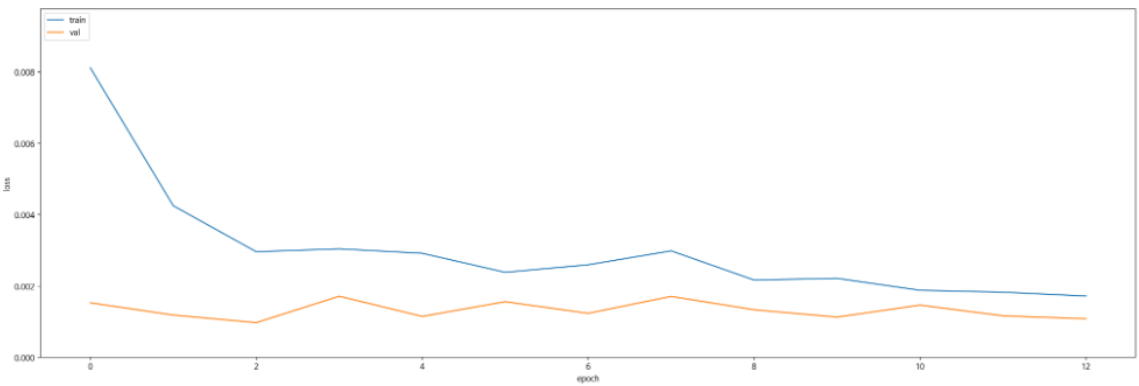
3. seq_length : 72 (12시간)

seq_length : 72

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

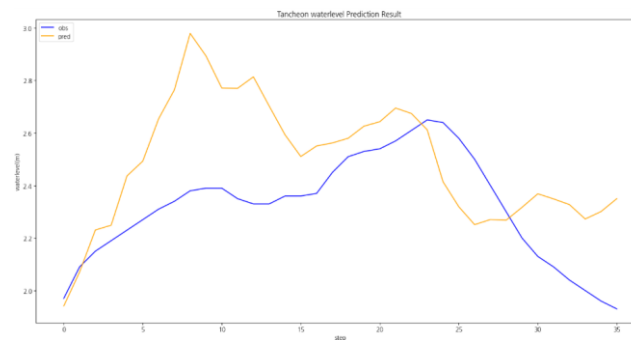
Train (3579 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1107	0.0626	0.2288	0.875847	0.877411	5.176867

Valid (878 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1056	0.0797	0.1431	0.868547	0.876040	18.826211

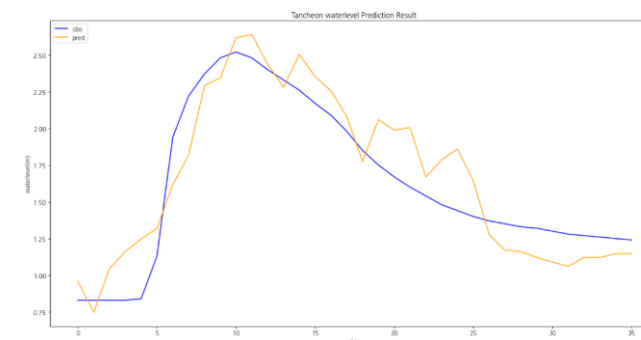
Test (378 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0922	0.0722	0.1278	0.840574	0.852845	21.710767

seq_length : 72
예측 코드 - 궁내교

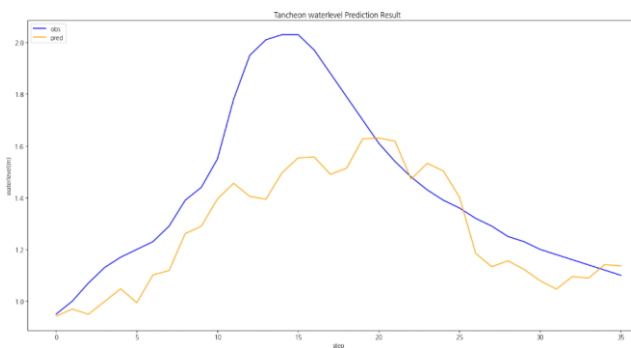
	궁내교
증설 전	testdata_220712 r2 : -5.784296501494748 mse : 0.26918999680221045 mae : 0.4563264531559414 rmse : 0.5188352308799109
	testdata_230711 r2 : 0.8198792444723559 mse : 0.05040516734425711 mae : 0.19690691431363427 rmse : 0.2245109515018301
증설 후	testdata_250716 r2 : 0.43715820892314816 mse : 0.056465539240365493 mae : 0.1745718653996785 rmse : 0.23762478667084688
	testdata_250814 r2 : 0.24260454014572064 mse : 0.05339795782693471 mae : 0.18680270228121015 rmse : 0.23107998145000513



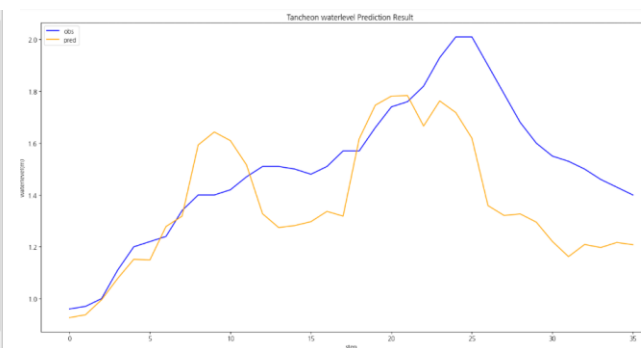
testdata_220712



testdata_230711



testdata_250716



testdata_250814

seq_length : 48 이 가장 좋음

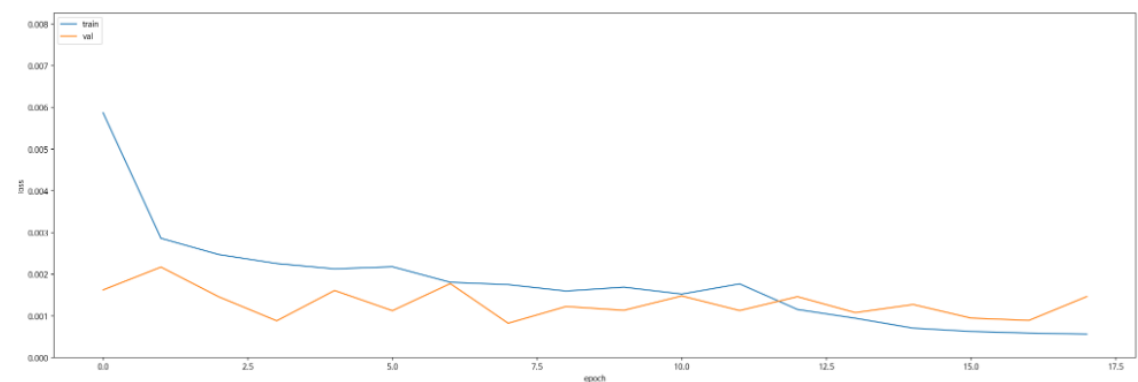
num_of_layer 조정
(LSTM 당 3개씩 말고)

0. 기존 num_of_layer : 12

num_of_layer =12
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

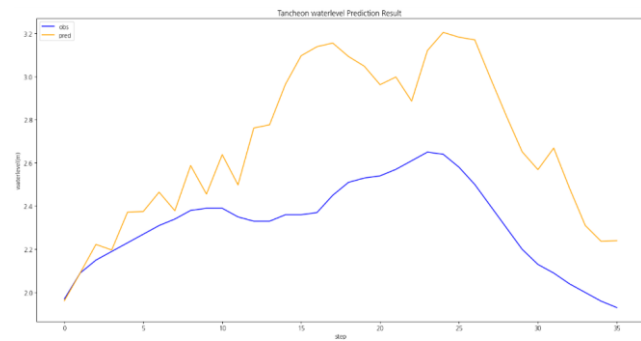
Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0929	0.0566	0.1681	0.916973	0.918180	32.452990

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0952	0.0712	0.1316	0.885653	0.887989	18.568189

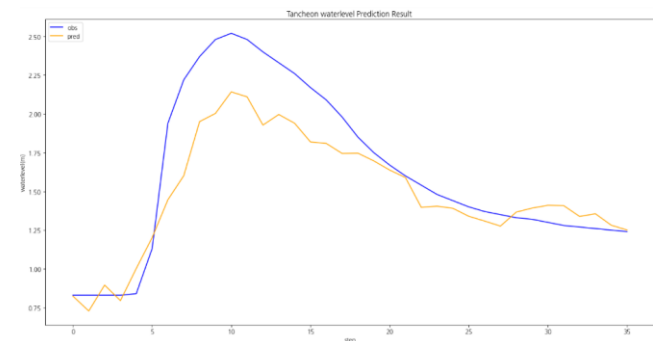
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0662	0.0498	0.0978	0.908553	0.917067	4.491322

num_of_layer =12
예측 코드 - 궁내교

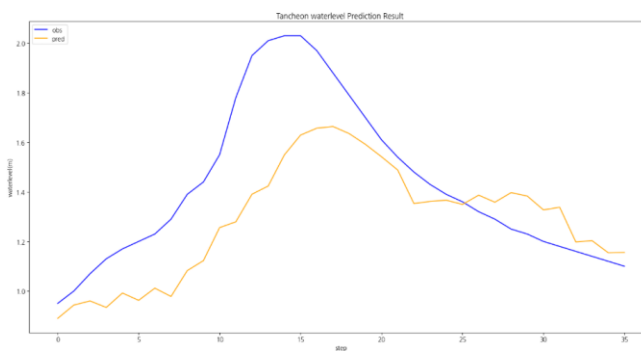
	궁내교
증설 전	testdata_220712 r2 : -3.7460557062787814 mse : 0.18831587329869987 mae : 0.37048792507913375 rmse : 0.43395376861907753
	testdata_230711 r2 : 0.7858148103033311 mse : 0.059937791720316914 mae : 0.17798842834101786 rmse : 0.24482195922816424
증설 후	testdata_250716 r2 : 0.3947407285994061 mse : 0.06072095512751068 mae : 0.19081864602035944 rmse : 0.24641622334479255
	testdata_250814 r2 : -0.056341420985954516 mse : 0.07447427089080348 mae : 0.2094912936952379 rmse : 0.27289974512777304



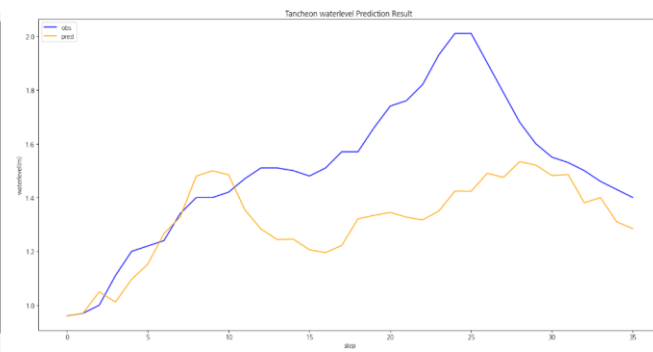
testdata_220712



testdata_230711



testdata_250716



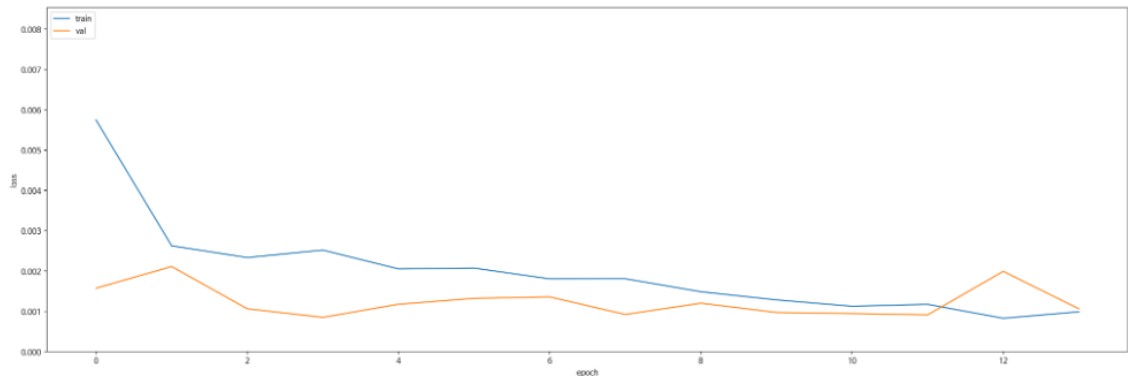
testdata_250814

1. num_of_layer =6

num_of_layer =6
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

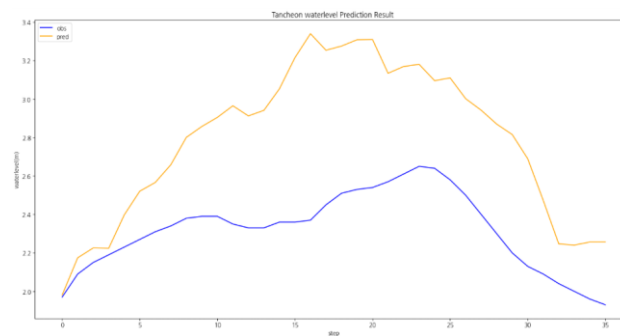
Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0983	0.0586	0.1902	0.893787	0.895242	1.412644

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1028	0.0801	0.1337	0.881882	0.901933	19.890367

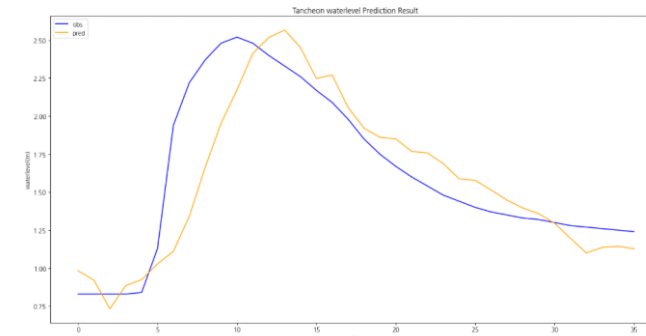
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0720	0.0568	0.0968	0.910301	0.916854	12.939607

num_of_layer = 6
예측 코드 - 궁내교

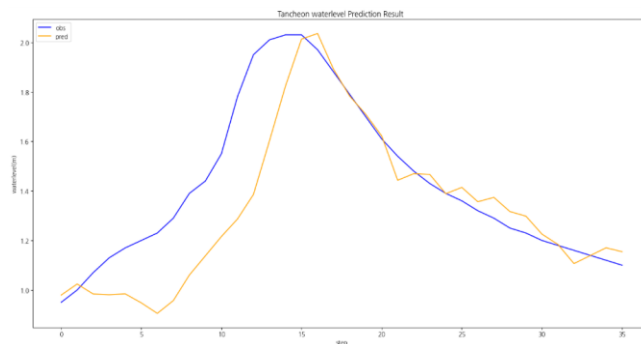
	궁내교
증설 전	testdata_220712 r2 : -6.008283559926514 mse : 0.27807744379538046 mae : 0.4701462470160591 rmse : 0.5273304882096052
	testdata_230711 r2 : 0.711225808358022 mse : 0.08081085054177745 mae : 0.19579490615261927 rmse : 0.2842724934667043
증설 후	testdata_250716 r2 : 0.592291528540601 mse : 0.040902219875632376 mae : 0.13294452905654908 rmse : 0.20224297237637795
	testdata_250814 r2 : 0.6348395741793602 mse : 0.025744570771242228 mae : 0.1266696121957567 rmse : 0.1604511476158467



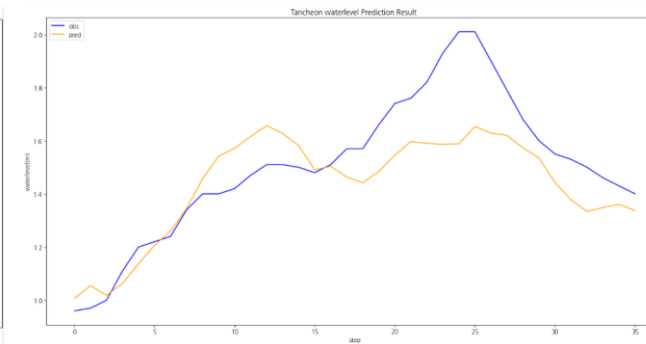
testdata_220712



testdata_230711



testdata_250716



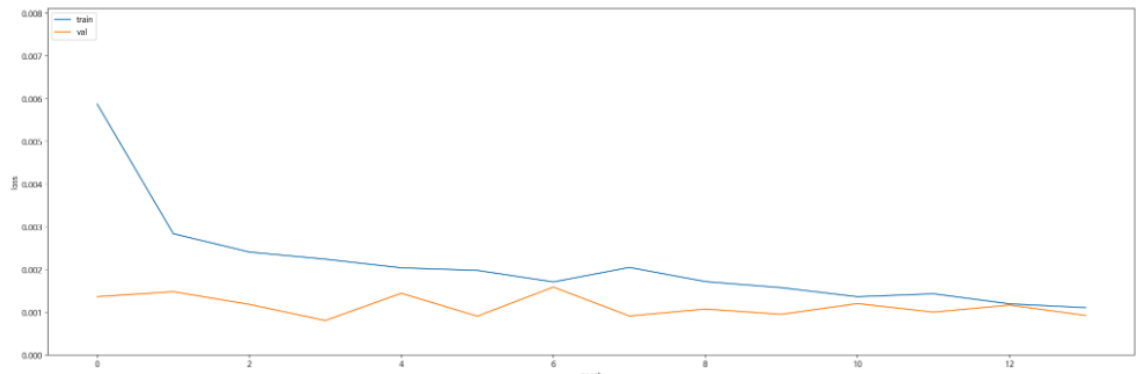
testdata_250814

2. num_of_layer = 18

num_of_layer =18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

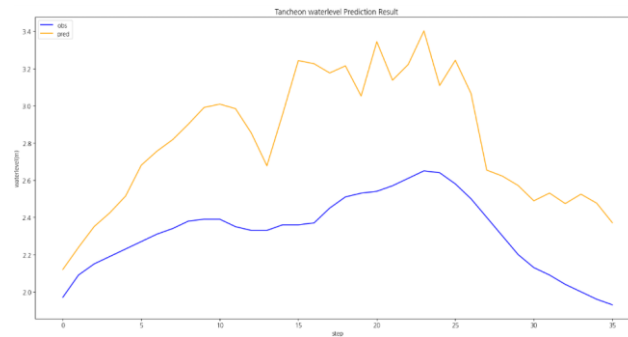
Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0949	0.0558	0.1886	0.895516	0.895592	4.164764

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0972	0.0750	0.1304	0.887686	0.896720	21.575676

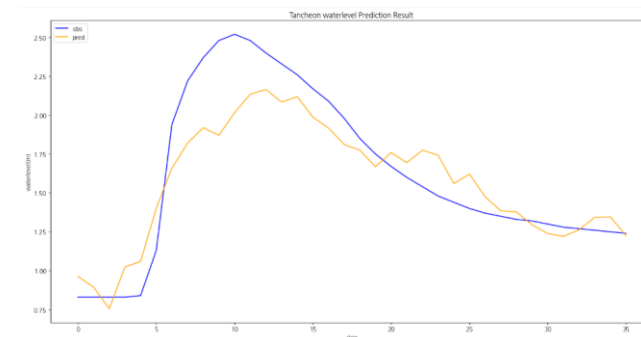
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0667	0.0518	0.0962	0.911507	0.914041	12.189970

num_of_layer =18
예측 코드 - 궁내교

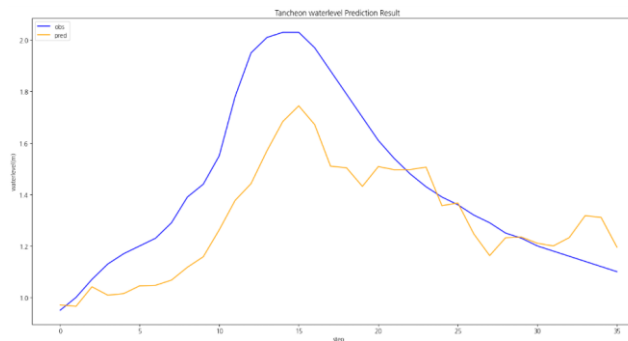
	궁내교
증설 전	testdata_220712 r2 : -6.167038119838788 mse : 0.2843765699414305 mae : 0.4995885904630026 rmse : 0.5332696971902965
	testdata_230711 r2 : 0.8158748707557312 mse : 0.05152575517825852 mae : 0.1779113178120719 rmse : 0.2269928527030279
증설 후	testdata_250716 r2 : 0.5277184502608925 mse : 0.04738033458438224 mae : 0.16776240130265554 rmse : 0.2176702427627218
	testdata_250814 r2 : 0.4561574682398356 mse : 0.038342031494366074 mae : 0.14596857958369786 rmse : 0.19581121391372372



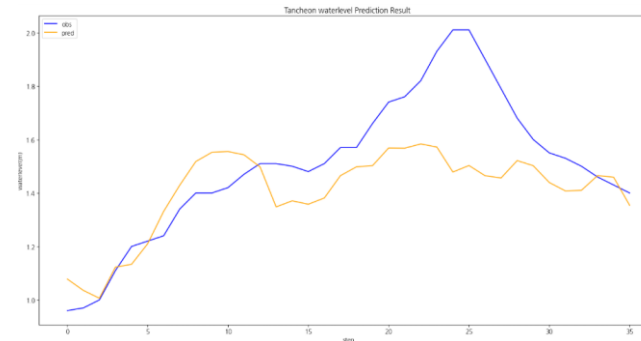
testdata_220712



testdata_230711



testdata_250716



testdata_250814

num_of_layer = 6 이 가장 낮은 성능
18 이 성능이 가장 좋음

영향 없는 관측소 및 컬럼 빼기

컬럼 정리 (+ MIET 학습 데이터 기본 설정)

궁내교

time	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	1.12	5.11	0.0

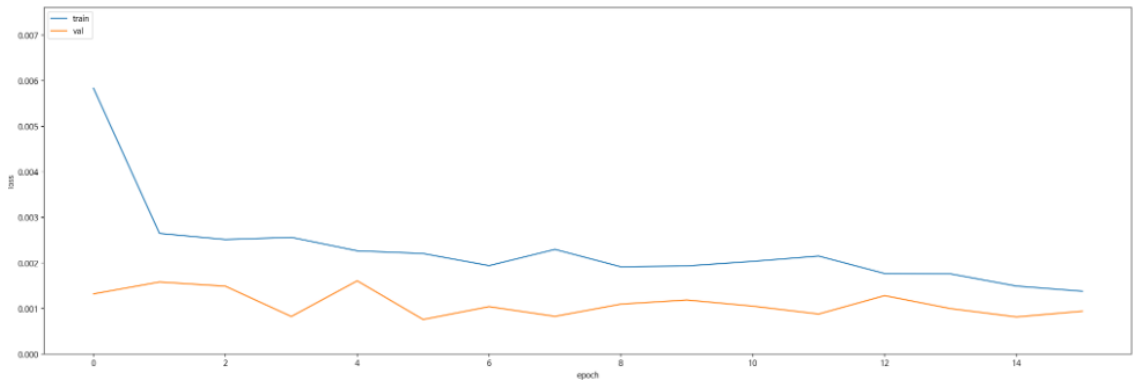
대곡교

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

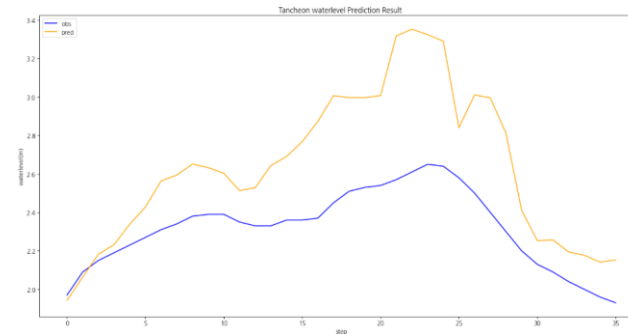
Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0992	0.0590	0.2025	0.879609	0.887010	2.961444

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0871	0.0633	0.1262	0.894740	0.899383	13.679203

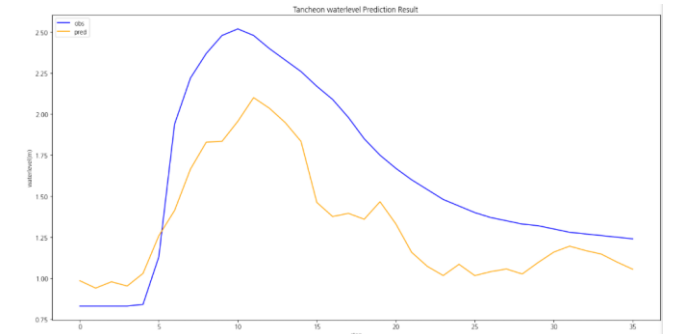
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0746	0.0548	0.1139	0.875795	0.896472	5.200029

예측 코드 - 궁내교

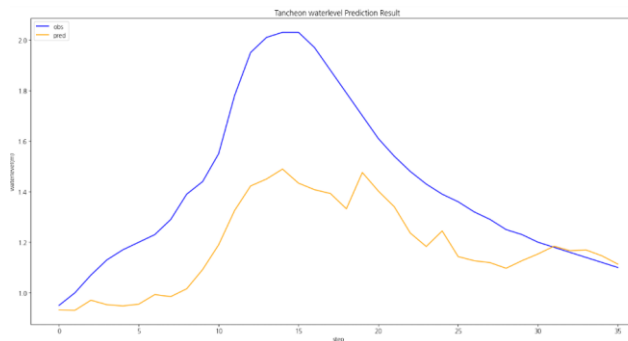
	궁내교
증설 전	testdata_220712 r2 : -2.6245451181948463 mse : 0.14381613311879415 mae : 0.3180192591084374 rmse : 0.37923097594842403
	testdata_230711 r2 : 0.45603772333825565 mse : 0.15222293235323595 mae : 0.34441678457789954 rmse : 0.39015757374839716
증설 후	testdata_250716 r2 : 0.07746982769549693 mse : 0.0925502769526373 mae : 0.2464591088559892 rmse : 0.3042207700875095
	testdata_250814 r2 : -0.0717979251904115 mse : 0.07556398663826813 mae : 0.2238872608873579 rmse : 0.27488904423106447



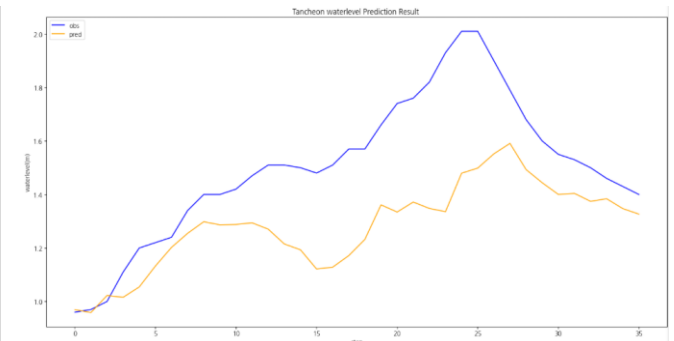
testdata_220712



testdata_230711



testdata_250716



testdata_250814

궁내교에 영향 없는 관측소
및 컬럼 빼기 성능 낮음

강우 이벤트 전부 학습으로 사용

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

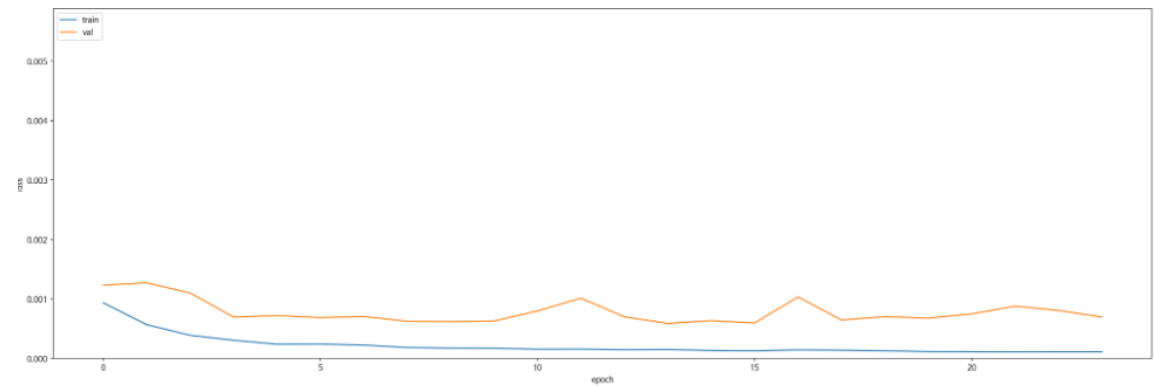
학습 : 총 1420개 中 1409개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

검증 : 학습 데이터에서 제외한 6개 데이터 사용

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 (?) 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

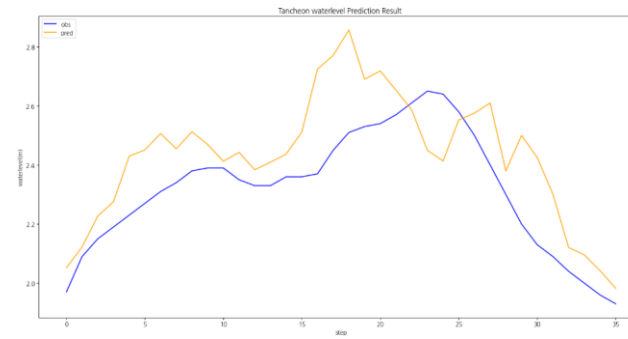
Train (23152 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0218	0.0209	0.0347	0.964980	0.967436	7.621150

Valid (6824 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0459	0.0488	0.0684	0.949009	0.961717	21.292681

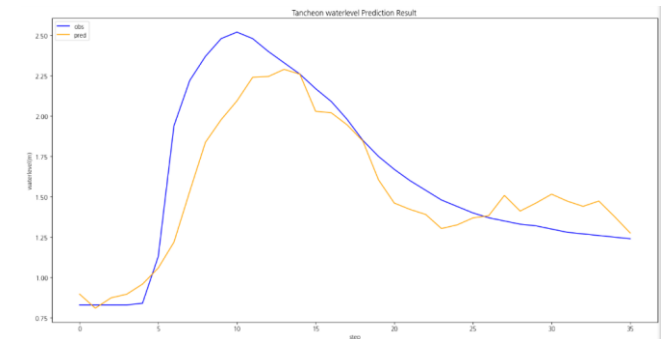
Test (5318 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0444	0.0528	0.0582	0.881387	0.924949	15.928333

예측 코드 - 궁내교

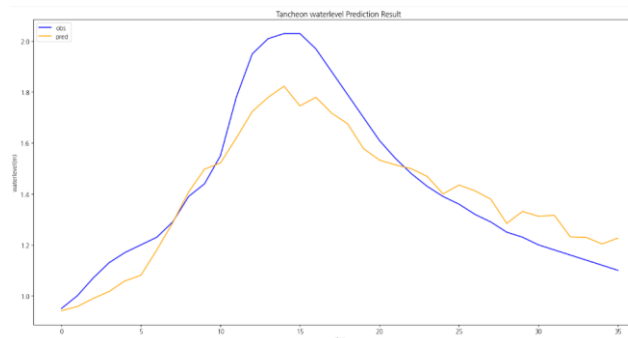
	궁내교
증설 전	testdata_220712 r2 : 0.2819461768473639 mse : 0.028491223270634754 mae : 0.1405930160151588 rmse : 0.16879343373080233
	testdata_230711 r2 : 0.7775425245477174 mse : 0.062252716208667874 mae : 0.17482565687762364 rmse : 0.24950494225299
증설 후	testdata_250716 r2 : 0.8592693835000157 mse : 0.014118408181981755 mae : 0.09755011134677463 rmse : 0.11882090801698898
	testdata_250814 r2 : 0.7044438615885542 mse : 0.020837323499961948 mae : 0.11466785464021895 rmse : 0.14435138897829125



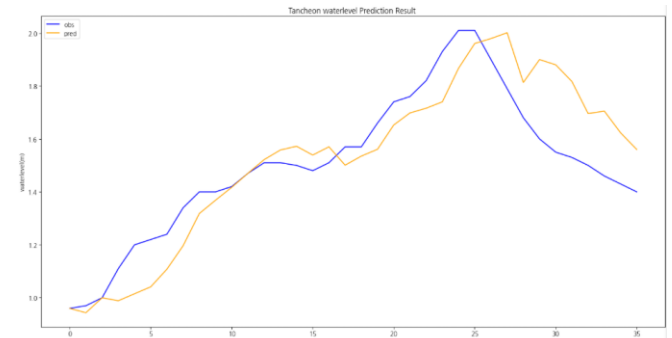
testdata_220712



testdata_230711



testdata_250716



testdata_250814

강우 이벤트 전부 학습으로 사용
 , num_of_layer 조정
(LSTM 당 3개씩 말고)

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1409개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

검증 : 학습 데이터에서 제외한 6개 데이터 사용

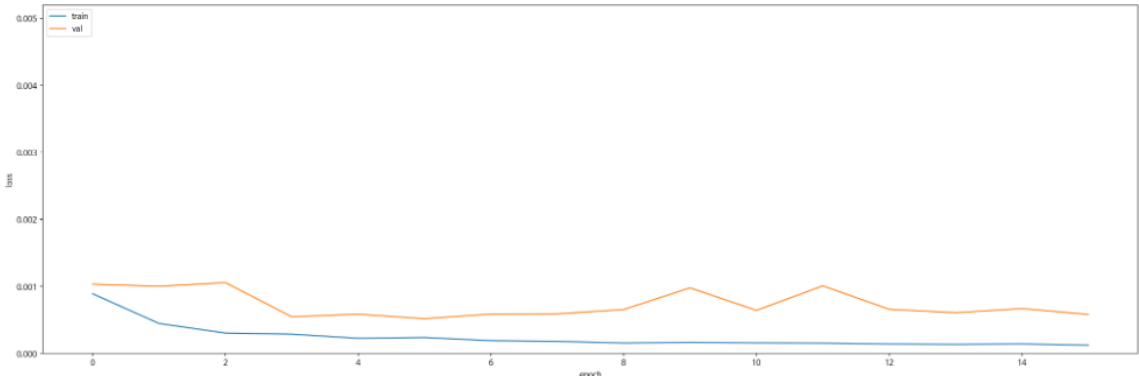
0. 기존 num_of_layer : 12

1. num_of_layer =6

num_of_layer =6
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 (?) 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

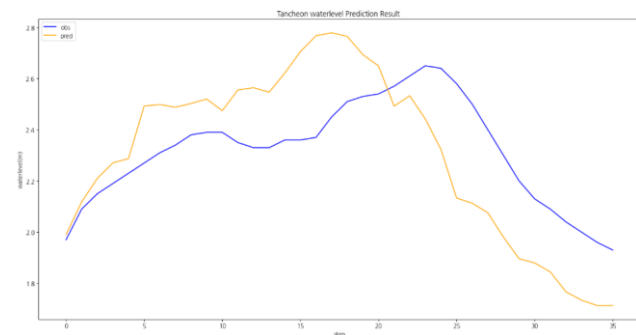
Train (23152 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0261	0.0253	0.0373	0.959507	0.962879	0.052707

Valid (6824 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0415	0.0425	0.0642	0.954954	0.964136	25.005565

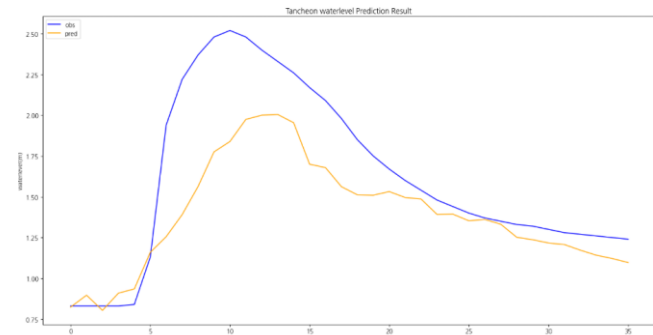
Test (5318 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0376	0.0443	0.0526	0.903120	0.925041	9.780254

num_of_layer = 6
예측 코드 - 궁내교

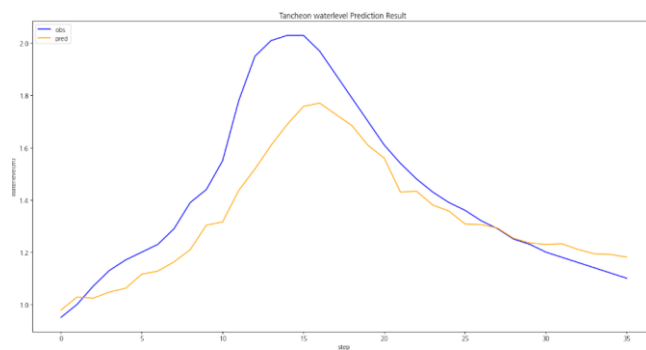
	궁내교
증설 전	testdata_220712 r2 : -0.4290769483646659 mse : 0.056703479730822445 mae : 0.21155922638045416 rmse : 0.23812492463163626
	testdata_230711 r2 : 0.5759262436072022 mse : 0.11867321228289035 mae : 0.24265570859114327 rmse : 0.3444897854550848
증설 후	testdata_250716 r2 : 0.7387643260557611 mse : 0.026207743333805916 mae : 0.1166555896732542 rmse : 0.16188805803333955
	testdata_250814 r2 : 0.8942565546130148 mse : 0.007455133198627011 mae : 0.05848213666015199 rmse : 0.08634311320902792



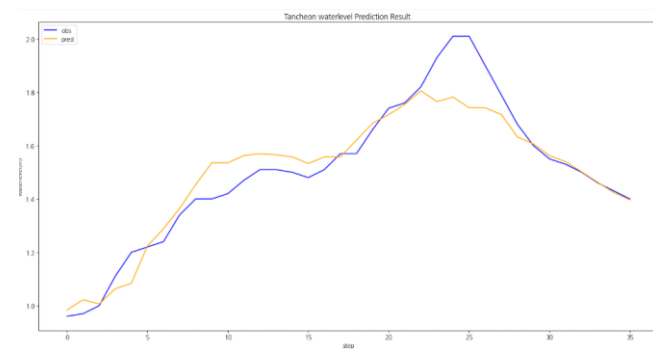
testdata_220712



testdata_230711



testdata_250716



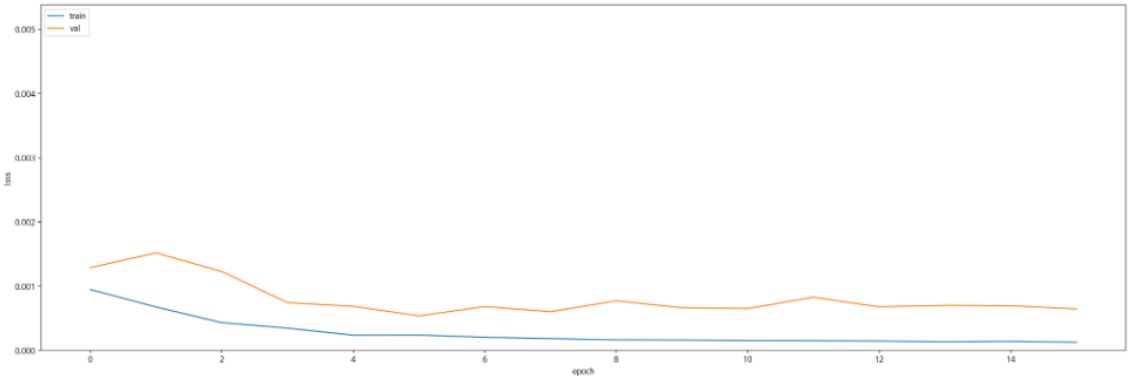
testdata_250814

2. num_of_layer = 18

num_of_layer =18
학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 (?) 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

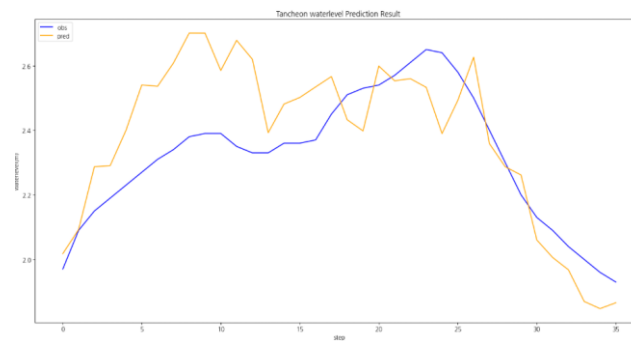
Train (23152 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0251	0.0241	0.0386	0.956557	0.957211	0.314478

Valid (6824 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0456	0.0483	0.0654	0.953377	0.964547	22.759481

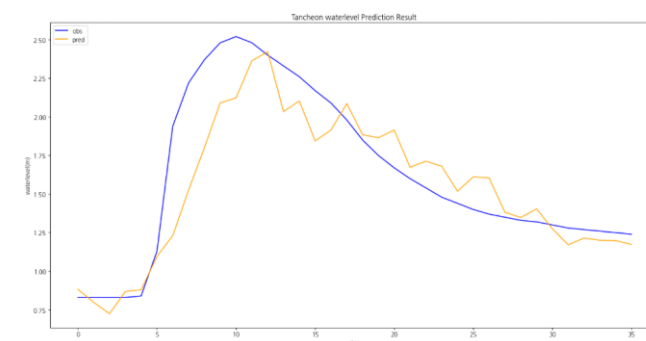
Test (5318 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0440	0.0521	0.0594	0.876632	0.917288	11.256301

num_of_layer =18
예측 코드 - 궁내교

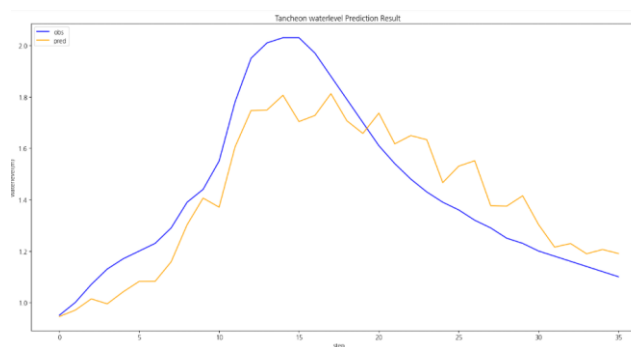
	궁내교
증설 전	testdata_220712 r2 : 0.3346054367651323 mse : 0.026401788351959294 mae : 0.13436633851793073 rmse : 0.16248627127225024
	testdata_230711 r2 : 0.7815566267629285 mse : 0.06112940594216386 mae : 0.1701013848516676 rmse : 0.24724361658526972
증설 후	testdata_250716 r2 : 0.78572611048365 mse : 0.021496432760479148 mae : 0.12670835547977022 rmse : 0.14661661829574146
	testdata_250814 r2 : 0.8621027839342135 mse : 0.009722041018504752 mae : 0.07590294844574397 rmse : 0.09860041084348864



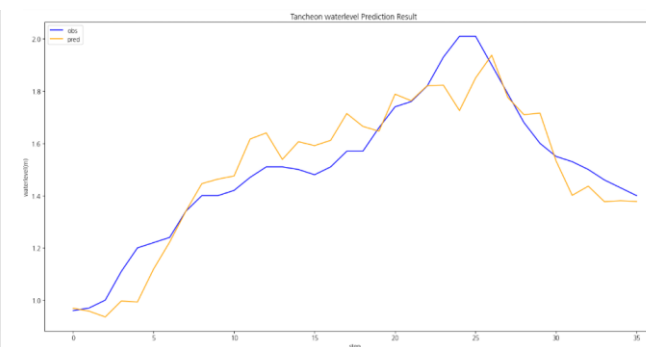
testdata_220712



testdata_230711



testdata_250716



testdata_250814

**num_of_layer = 6 을 제외하고
12 18 은 비슷한 성능**

강우 이벤트 전부 학습으로 사용,
영향 없는 관측소 및 컬럼 빼기

MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 12 (각 LSTM 3개 처리)

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1257개 中 1246개 데이터 파일 학습데이터로 사용 (결측 데이터 5개)

검증 : 학습 데이터에서 제외한 6개 데이터 사용

컬럼 정리 (+ MIET 학습 데이터 기본 설정)

궁내교

time	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	1.12	5.11	0.0

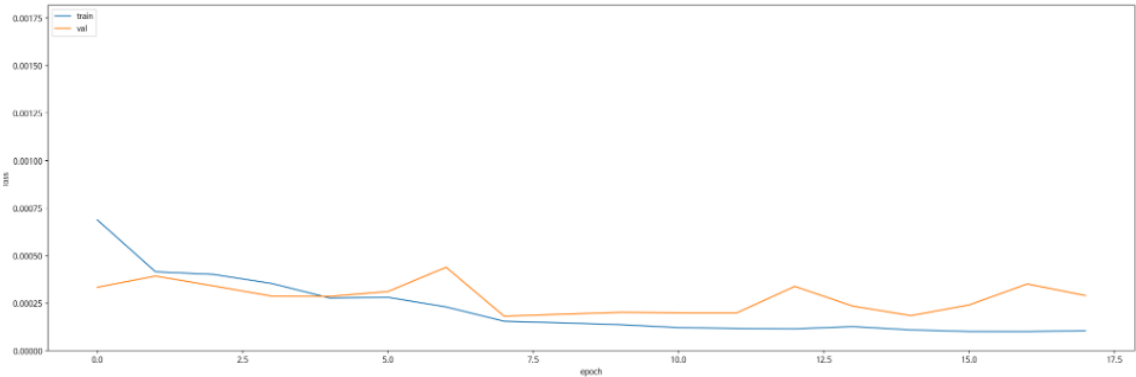
대곡교

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 (?) 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

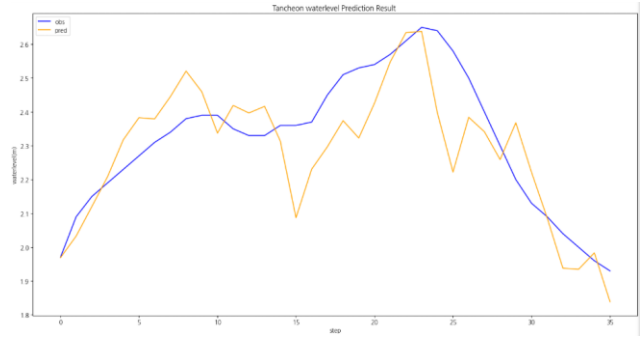
Train (19934 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0311	0.0292	0.0485	0.947363	0.949078	14.194829

Valid (5933 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0382	0.0425	0.0547	0.932596	0.941128	8.046630

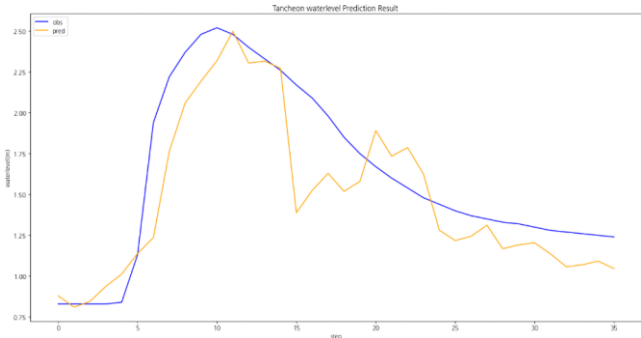
Test (4526 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0398	0.0452	0.0565	0.899546	0.915580	4.214151

예측 코드 - 궁내교

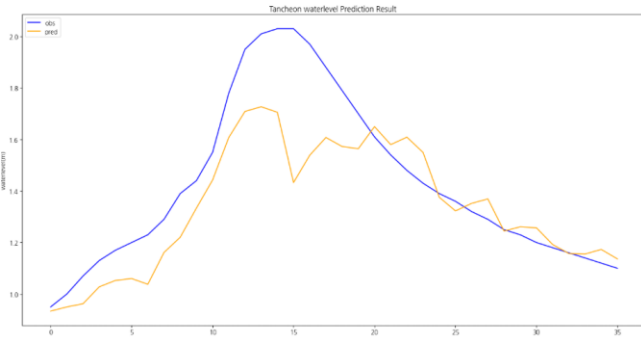
궁내교	
증설 전	testdata_220712 r2 : 0.6183725611506388 mse : 0.015142364285060553 mae : 0.0959299246470133 rmse : 0.12305431436995841 testdata_230711 r2 : 0.7409383899519266 mse : 0.07249605282130708 mae : 0.20018566151460013 rmse : 0.2692509105301356
증설 후	testdata_250716 r2 : 0.6758801545804565 mse : 0.032516423158811765 mae : 0.1282569737566842 rmse : 0.18032310766735296 testdata_250814 r2 : 0.5985093638956797 mse : 0.02830592628417979 mae : 0.1281632967789968 rmse : 0.1682436515419818



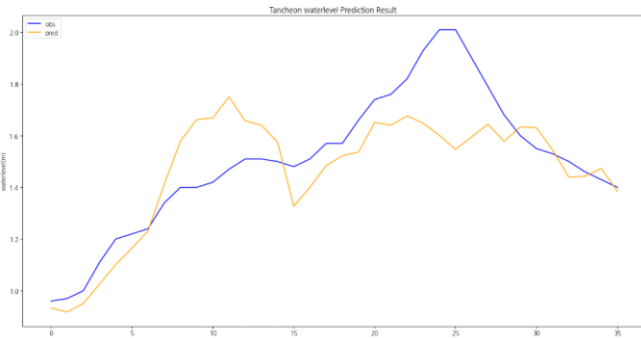
testdata_220712



testdata_230711



testdata_250716



testdata_250814